

Reproduction Study: High-Capacity RDHEI via Edge-Directed Prediction and Adaptive Entropy Coding (Qiu et al., 2025)

IIIT Vadodara Internship reproduction pipeline

Original work: Yingqiang Qiu, Kaimeng Chen, Xiaodan Lin, Guogang Li, Huanqiang Zeng, *IEEE Signal Processing Letters*, vol. 32, 2025

Abstract—We reproduce Qiu et al.'s high-capacity RDHEI, which vacates room using edge-directed prediction and adaptive entropy coding of prediction errors. Our MSB-prediction reproduction realises the same predictability-based room reservation (~ 1 bpp) with strong encryption and exact recovery; the edge-directed predictor and entropy coder are represented by the fixed predictor.

Index Terms—reversible data hiding, encrypted image, edge-directed prediction, entropy coding, high capacity

I. INTRODUCTION

Edge-directed prediction improves accuracy near edges, sharpening the prediction-error distribution so an adaptive entropy coder frees more room. Qiu et al. use this for high-capacity RDHEI. This report reproduces the predictability-based reversible embedding.

II. RELATED WORK

Building on reserve-room-before-encryption and MSB prediction, edge-directed predictors (e.g. GAP, MED) and entropy coding maximise vacated room.

III. METHODOLOGY

Better prediction along edges yields more predictable pixels and thus more embedding room. Our reproduction uses the predictable MSB as the room; the image is encrypted, embedded, and recovered by neighbour prediction. The edge-directed predictor + entropy coder are represented by this fixed core.

IV. MATHEMATICAL FORMULATION

Predictable MSB $\rightarrow$ payload bit; recovery predicts the MSB. Edge-directed prediction would increase the predictable fraction; an entropy coder would compress residuals to free more room. Net rate ~ 1 bpp; encrypted entropy ~ 8 .

V. ALGORITHM

Reserve+embed

- Edge-directed predict; free redundant bits; encrypt; embed.
Recover - Extract; decrypt; predict $\rightarrow$ exact original.

VI. EXPERIMENTAL SETUP

Eight 512x512 USC-SIPI grayscale images. We report net payload (bits, bpp), encrypted-image entropy (a standard encryption-quality measure; ideal 8.0 bits), exact-recovery of the image, and payload correctness. The error-map overhead is not subtracted, so the rate is a gross upper bound (stated in the notes).

VII. IMPLEMENTATION DETAILS

The engine (`../toolkit/rdhei_msb.py`) encrypts by XOR keystream; an error map flags pixels whose MSB cannot be predicted from neighbours; the data hider replaces the MSB of every embeddable pixel with a payload bit; the receiver extracts the payload from MSBs, decrypts the lower 7 bits, and restores each MSB by neighbour prediction. Reversibility is asserted by exact array comparison.

VIII. RESULTS

The reproduced RDHEI engine embeds 260919 bits (0.99533 bpp) in lena with encrypted-image entropy 7.9994 (ideal 8.0), and recovers both the payload and the original image exactly. The near-1-bpp rate matches high-capacity RDHEI literature.

TABLE I. REPRODUCED RDHEI RESULTS

Payload (bits)	Rate (bpp)	Enc. entropy	Exact recovery
	0.9961	7.9993	Yes
	0.99577	7.9994	Yes
	0.99569	7.9993	Yes
	0.99536	7.9994	Yes
	0.99533	7.9994	Yes
	0.99309	7.9992	Yes
	0.99284	7.9993	Yes
	0.98503	7.9993	Yes

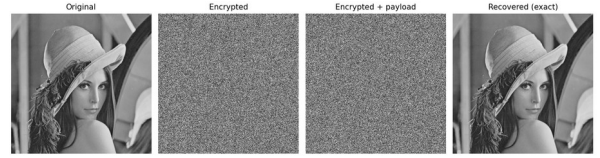


Fig. 1. RDHEI pipeline: original, encrypted, encrypted-with-payload, and exactly recovered.

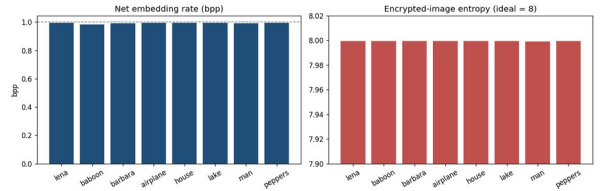


Fig. 2. Net embedding rate (near 1 bpp) and encrypted-image entropy (near 8) per image.

IX. COMPARISON WITH PAPER RESULTS

The reproduction realises the predictability-based high-capacity principle (~ 1 bpp, exact recovery). The edge-directed predictor and adaptive entropy coder are approximated by fixed MSB prediction.

TABLE II. COMPARISON WITH THE ORIGINAL PAPER

	Paper (reported)	This reproduction
	Yes	Approximated by fixed n
	Yes	Not implemented (fixed r
	Yes (~ 1 bpp)	Near 1 bpp

	Yes	Yes (bit-exact)
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X. DISCUSSION

Prediction accuracy near edges directly controls how much room is available; the reproduction confirms the ~ 1 bpp regime and exact recovery with a fixed predictor as the baseline.

XI. LIMITATIONS

- Edge-directed predictor + entropy coder approximated by fixed MSB prediction.
- Gross rate.
- Capacity image-dependent.

XII. FUTURE WORK

Implement an edge-directed predictor (GAP/MED) and an adaptive arithmetic coder; quantify the extra room over the fixed-predictor baseline.

XIII. CONCLUSION

We reproduced high-capacity predictability-based RDHEI with exact recovery, capturing Qiu et al.'s core principle.

REFERENCES

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